Bihar Engineering University, Patna

B.Tech 4th Semester Examination, 2024

Course : B.Tech
Code : 101404

Subject: Introduction to Fluid Mechanics

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer of the following (any seven question only): **[2 x 7 = 14]**

(a) Kinematic Viscosity is defined as
(i) Dynamic viscosity x density
(ii) Dynamic viscosity/density
(iii) Dynamic viscosity x pressure
(iv) Pressure x density

(b) Stoke is unit of
(i) Surface tension
(ii) Viscosity
(iii)Kinematic Viscosity
(iv) None of above

(c) Coefficient of discharge (C_d) in terms of C_v and C_c is
(i) $C_d = C_c \times C_v$
(ii) $C_d = C_v / C_c$
(iii) $C_d = C_c / C_v$
(iv) None of above

(d) The value of the kinematic energy correction factor (α) of viscous flow through a circular pipe is
(i)1.33
(ii) 1.50
(iii)2.0
(iv)1.25

(e) Continuity equation deals with the law of conservation of
(i) Mass
(ii) Momentum
(iii)Energy
(iv) None of above

(f) Rynold's number is defined as the
(i)Ratio of inertia force to gravity force
(ii) Ratio of viscous force to gravity force
(iii) Ratio of viscous force to Elastic force
(iv) Ratio of inertia force to viscous force.

(g) The dividing factor for converting one poise to MKS unit of dynamic viscosity is
(i)9.81
(ii) 98.1
(iii) 981
(iv) 0.981

(h) Atmospheric pressure held in terms of water column is
(i) 7.5m
(ii) 8.5m
(iii)9.81m
(iv) 10.30m

(i) Pitot tube is used for measurement of
(i) Pressure
(ii) flow
(iii) velocity at a point
(iv) Discharge

(j) Boundary layer on flat plat is called as laminar boundary layer if
(i) Renould number <2000
(ii) Renould number <4000
(iii) Renould number $<5 \times 10^5$
(iv) None of above

© Apna BEU
ORIGINAL DOCUMENT - EDC - 107 @ 27-02-2025 09:31:55

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play

Q.2 (a) Define Following terms with SI Units: i) Weight density ii) Dynamic Viscosity iii) Bulk Modulus iv) Capillarity. [7]

(b) The space between the two square flat parallel plates is filled with oil. Each side of plate is 60cm. the thickness of the oil film is 12.5mm. the upper plates which moves with velocity 2.5m/s requires a force of 98.1N to maintain the speed. Determine the viscosity of the oil in poise. Also find the kinematic viscosity of the oil in stokes. If the specific gravity of oil is 0.95. [7]

Q.3 (a) Petrol of specific gravity 0.8 flaws upward through a vertical pipe. A & B are two points in the pipes, B being 0.3m higher than A. connections are led from A&B to a U tube containing mercury. If the difference of pressure Between A & B is 0.18kgf/cm^2. Find the difference in the mercury level in the differential manometer. [7]

(b) Derive an Expression for total pressure force and depth of pressure for a vertical surface submerged in water. [7]

Q.4 (a) What is a venturi meter? Derive an expression for the discharge through a venturi meter. [7]

(b) An orifice meter with orifice diameter 15 cm is inserted in a pipe of 30 cm diameter. The pressure difference measured by a mercury oil differential manometer on the two sides of the orifice meter gives a reading of 50 cm of mercury. Find the flow rate of flow of oil of sp. gr. 0.9 when the co-efficient of discharge of the orifice meter=0.64. [7]

Q.5 (a) Distinguish between: i) Ideal fluid and real fluid ii) Manometer and pressure gauge iii) Surface tension and capillarity. [7]

(b) State and proof pascal's law. [7]

Q.6 (a) Differentiate between i) Orifice and Mouthpiece ii) Notch and Wier [4]

(b) A compound pipes in series consists of 1800m long of 0.5m diameter, 1200m long of 0.4m diameter and 600m long of 0.3m diameter connected between two tanks with difference in water level of 100m. Determine the flow rate in the pipe neglecting minor losses. Also compute the diameter of equivalent pipe to be connected between two tanks. Take f=.0.04 in $h_f=flv^2/ZgD$. [10]

Q.7 (a) Differentiate between i) Uniform and non-uniform flow ii) steady and unsteady flow. [4]

(b) Derive continuity equation for a three-dimensional flow Cartesian coordinate. [10]

Q.8 (a) Obtain an expression for Euler's equation of motion along a stream line and deduce it to Bernoulli's equation. [8]

(b) Show that streamlines and equipotential lines form a set of perpendicular lines. [6]

Q.9 (a) If 10000 Liters of certain liquid weigh 1329kN. Calculate. [7]
i) Specific weight ii) Mass density iii) Specific volume iv) Specific gravity.

(b) Derive the expression for discharge through a rectangular notch. [7]

© Apna BEU

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play